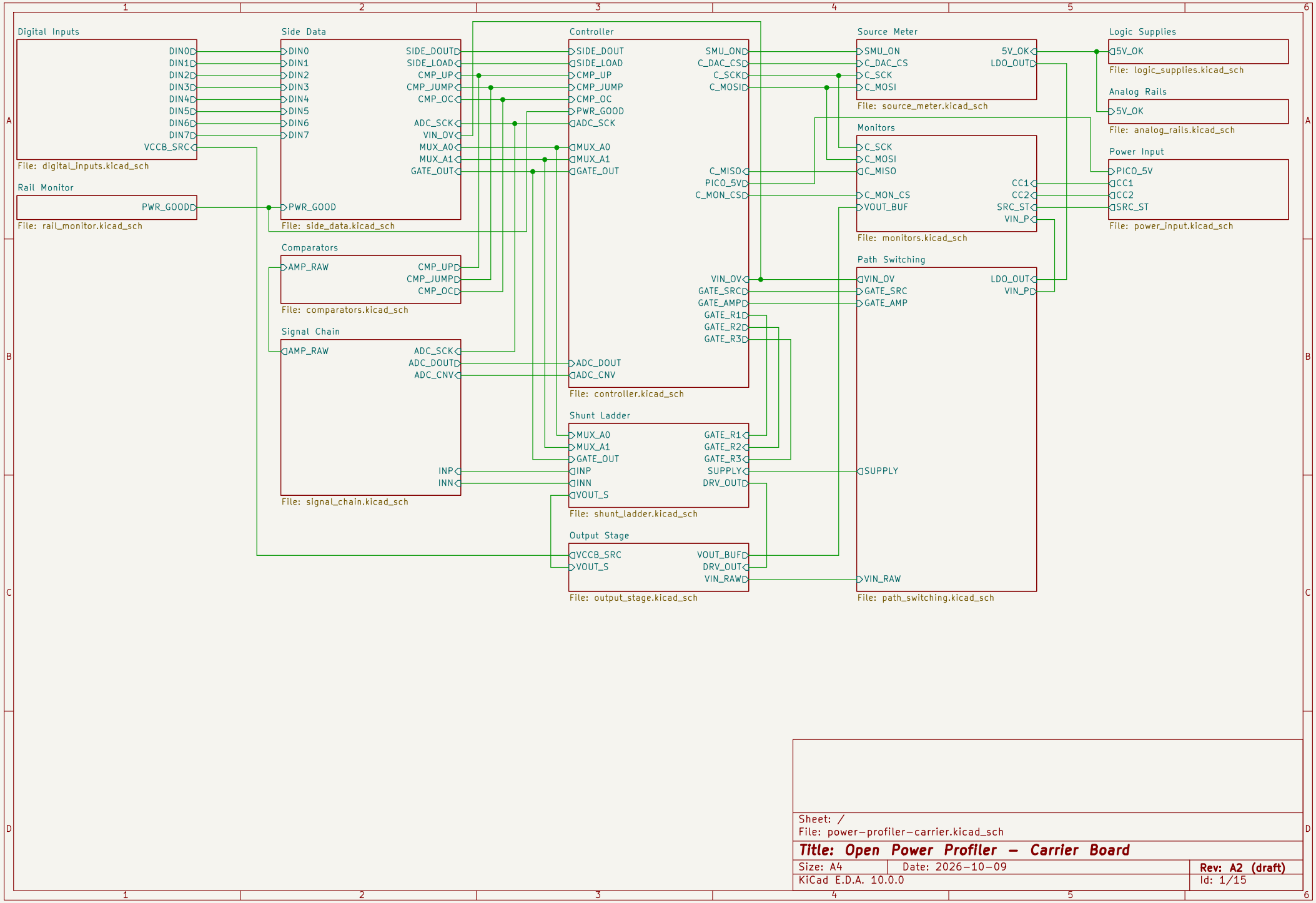


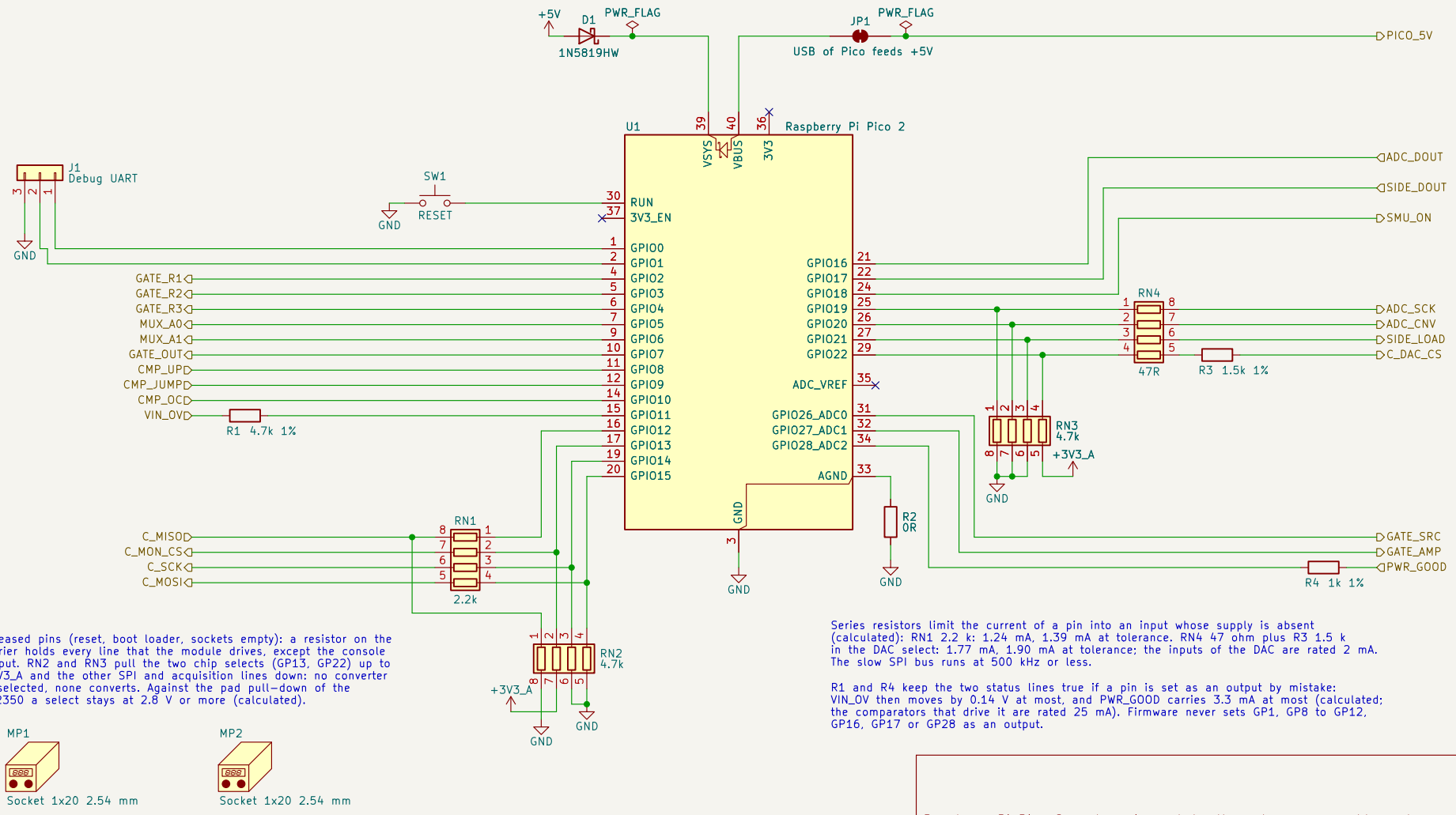
Sheet: /
File: power-profiler-carrier.kicad_sch

Title: Open Power Profiler – Carrier Board

Size: A4	Date: 2026-10-09	**Rev: A2 (draft)**
KiCad E.D.A. 10.0.0		Id: 1/15

U1 is a Raspberry Pi Pico 2 with headers, plugged into the sockets MP1 and MP2: the only programmable part. RP2350 stepping A3 or A4 is preferred. A Pico 2 W fits the sockets, but the firmware does not support it.
PIO block 0 (GP16, GP17, GP19 to GP21) clocks the converter and reads its data and the side data in the same word. PIO block 1 (GP2 to GP10) is the range sequencer: comparators in, gates and multiplexer address out. J1: TX, RX, GND at 3.3 V.

Supply: the USB voltage of the module leaves through JP1 as PICO_5V to its current limiter (0.76 A) and to the source selector on the Power Input sheet, where USB-C has priority. The selected +5V feeds VSYS through D1. Cut JP1 to keep the carrier off the USB port of the computer.



Released pins (reset, boot loader, sockets empty): a resistor on the carrier holds every line that the module drives, except the console output. RN2 and RN3 pull the two chip selects (GP13, GP22) up to +3V3_A and the other SPI and acquisition lines down: no converter is selected, none converts. Against the pad pull-down of the RP2350 a select stays at 2.8 V or more (calculated).

Series resistors limit the current of a pin into an input whose supply is absent (calculated): RN1 2.2 k: 1.24 mA, 1.39 mA at tolerance. RN4 47 ohm plus R3 1.5 k in the DAC select: 1.77 mA, 1.90 mA at tolerance; the inputs of the DAC are rated 2 mA. The slow SPI bus runs at 500 kHz or less.

R1 and R4 keep the two status lines true if a pin is set as an output by mistake: VIN_OV then moves by 0.14 V at most, and PWR_GOOD carries 3.3 mA at most (calculated; the comparators that drive it are rated 25 mA). Firmware never sets GP1, GP8 to GP12, GP16, GP17 or GP28 as an output.

Raspberry Pi Pico 2 on two pin sockets: the only programmable part

Sheet: /Controller/
File: controller.kicad_sch

Title: Controller: Raspberry Pi Pico 2

Size: A4 | Date: 2026-10-09 | **Rev: A2 (draft)**
KiCad E.D.A. 10.0.0 | Id: 2/15

J2 is a USB-C sink for 5 V. R7 and R11 pull CC1 and CC2 down; the firmware reads the CC voltage through the monitor converter: below 0.61 V default USB, 0.70 V to 1.16 V 1.5 A, 1.31 V to 2.04 V 3 A. U2 protects the CC pins; D2, C2 and C3 sit at the receptacle, ahead of the current limit. D2 stands off 10 V; the input tolerates no more than that.
U4 (eFuse A) turns on when VBUS_C is above 2.92 V to 3.09 V and off below 2.67 V to 2.87 V (R9, R10, which also bleed VBUS_C). It limits the current to 2.0 A (1.81 A to 2.17 A, R15) and clamps its output to 5.28 V to 5.61 V when the input is above 5.54 V to 5.83 V (R12). C4 sets a turn-on delay of about 0.25 ms and a ramp of 13 V/ms (datasheet, typical). TP3 shows the input current, 0.297 V per ampere, behind R16.
R20, C9 and C10 damp the ring of a contact that closes; D4 takes their charge back to the input when the plug is pulled, so that the output stays within 0.3 V of the input.

U3 (eFuse B) does the same for the supply from the USB connector of the controller module: same thresholds (R5, R6), limit 0.76 A (0.67 A to 0.85 A, R13), fastest ramp, turn-on delay about 0.08 ms typical. R17 defines VBUS_P while U3 is off.
R14 and C5 (0.33 ohm, 10 uF) damp VBUS_P when the data cable is plugged again while USB-C feeds +5V: without them VBUS_P reaches 5.94 V (simulated); U5 is rated 6.0 V (datasheet).
U5 drives +5V from USB-C (VBUS_CP) while PR1 is above 1.0 V, that is while VBUS_CP is above 2.37 V (2.15 V to 2.59 V, calculated from R18, R19), otherwise from the module (VBUS_P). It blocks reverse current into both inputs. C8 delays PR1 by 0.19 ms, until the ramp of U4 is nearly complete.
Order of the thresholds on a falling rail: supervisor of +5V 3.83 V to 4.00 V (next sheet), eFuse off 2.67 V to 2.87 V, change of source 2.15 V to 2.59 V. The multiplexer leaves USB-C only when U4 is already off.
SRC_ST (TP5): open = USB-C in use, low = supply of the module in use.

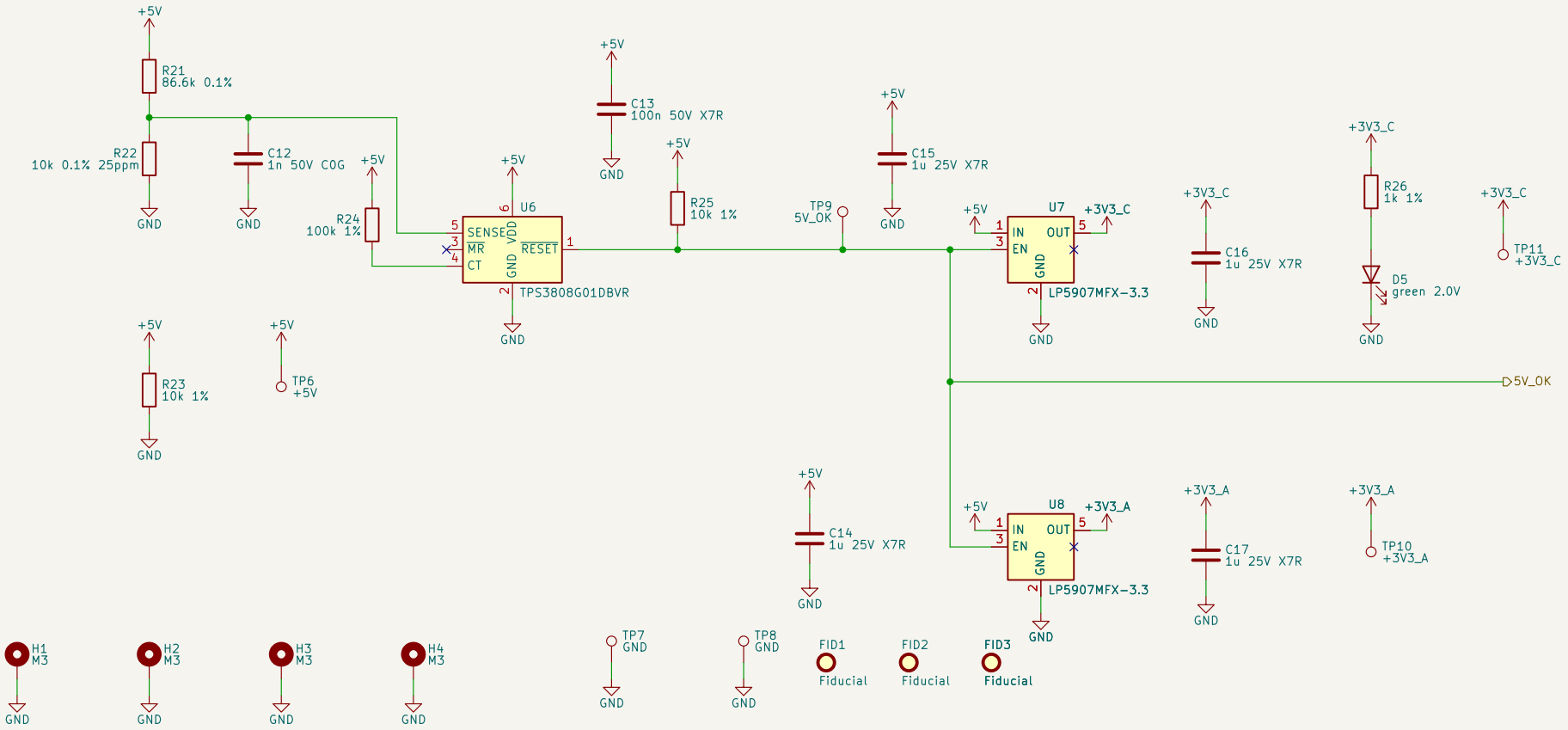
J2
USB-C power
VBUS
A9
CC1
A5
CC2
B5
SHIELD
GND
SH
A12
TP1
VBUS_C
D2
SMAJ10A
C2
4.7u 25V X7R
C3
100n 50V X7R
R9
4.53k 1%
R10
3.01k 1%
R12
402k 1%
C4
3.3n 50V X7R
PWR_FLAG
D4 1N5819HW
TP2
VBUS_CP
U4
VIN
OUT
EN/UVLO
FLT
TPS259621DDAR
OVLO/OVCSEL
dV/dT
ILM
C7
1u 25V X7R
R16
10k 1%
TP3
IMON
R15
453R 1%
R20
0.33R 1%
C9
22u 10V X7R
C10
22u 10V X7R
U2
TPD4E1U06DBVR
IO1
IO2
IO3
IO4
R7
5.1k 1%
R11
5.1k 1%
CC2
CC1
U5
VIN1
VIN2
PR1
MODE
VOUT
ST
TPS2116DRLR
+5V
C11
47u 16V
R18
13.7k 1%
R19
10k 1%
C8
33n 50V X7R
TP4
VBUS_P
TP5
SRC
SRC_ST
D3
1N5819HW
U3
PICO_5V
C1
100n 50V X7R
R5
4.53k 1%
R6
3.01k 1%
R8
402k 1%
R14
0.33R 1%
C5
10u 25V X5R
C6
1u 25V X7R
R17
100k 1%
R13
1.21k 1%

Two USB inputs, one eFuse each, joined by a priority multiplexer that drives +5V

Sheet: /Power Input/
File: power_input.kicad_sch

Title: Power Input: USB-C, Current Limits and Source Selection

Size: A4	Date: 2026-10-09	Rev: A2 (draft)
KiCad E.D.A. 10.0.0		Id: 3/15



U6 watches the 5 V rail. 5V_OK goes low when +5V falls below 0.405 V x (1 + R21 / R22) = 3.91 V (3.83 V to 4.00 V with the tolerances) and returns 0.18 s to 0.42 s after the rail has risen above 4.12 V at most (R24 selects 0.3 s typical). 5V_OK is open drain with R25 to +5V. Low, it switches off U7 and U8 here and, on other sheets, the +12 V regulator, the charge pump and the pre-regulator: the carrier has one off state, in which only +5V and +13V5 stay up. C12 filters SENSE with 9 us.
TP9 held to ground keeps the carrier off during bring-up.

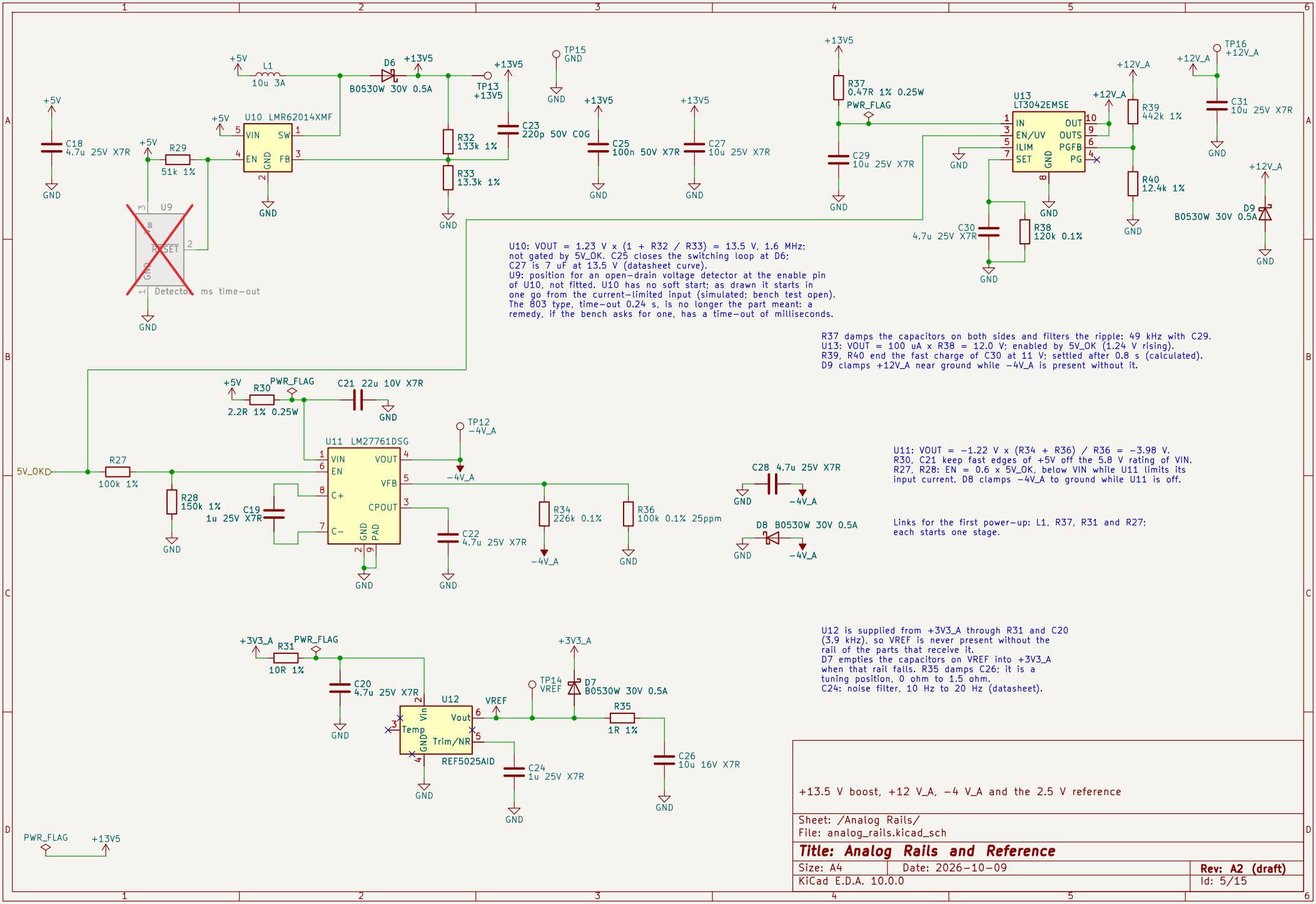
U7: +3V3_C, logic of the carrier. U8: +3V3_A, ADC, comparators, DAC, monitor converter. The regulators are specified for 10 uF at most on the output (datasheet). Over all sheets +3V3_A holds 3.8 uF and 4.7 uF more behind the 10 ohm of the reference supply, +3V3_C 2.4 uF (calculated). A disabled regulator empties its output through 230 ohm (datasheet), so +3V3_A starts from 0 V after every dip of the rail.
R26 and D5: power indicator, 1.3 mA. R23 bleeds +5V when no source is connected.
H1 to H4: mounting holes, tied to ground. FID1 to FID3: fiducials for the placement machine.

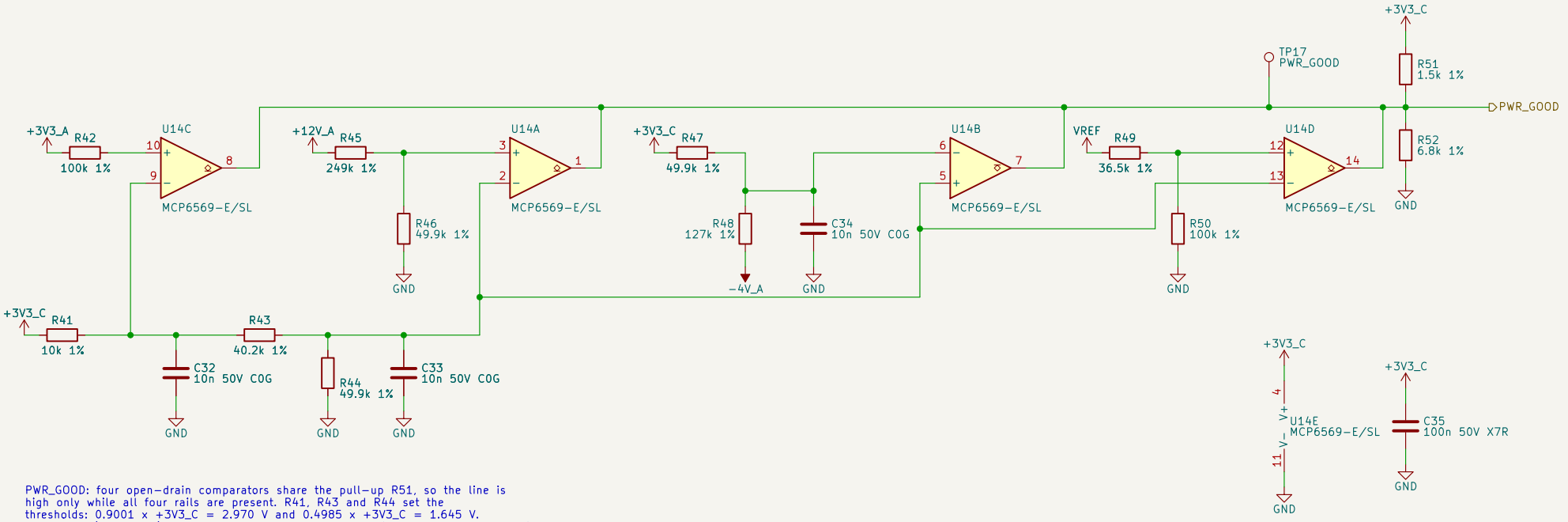
Supervisor of the 5 V rail; its output 5V_OK enables the two 3.3 V regulators

Sheet: /Logic Supplies/
File: logic_supplies.kicad_sch

Title: Logic Supplies and Rail Supervisor

Size: A4	Date: 2026-10-09	**Rev: A2 (draft)**
KiCad E.D.A. 10.0.0		Id: 4/15


+5V
L1
10u 3A
D6
B0530W 30V 0.5A
+13V5
TP13
+13V5
TP15
GND
C18
4.7u 25V X7R
R29
51k 1%
U10 LMR62014XMF
VIN
SW
EN
GND
FB
R32
133k 1%
R33
13.3k 1%
C23
220p 50V C0G
C25
100n 50V X7R
C27
10u 25V X7R
U9
RESET
Detector
ms time-out
U10: VOUT = 1.23 V x (1 + R32 / R33) = 13.5 V, 1.6 MHz;
not gated by 5V_OK. C25 closes the switching loop at D6;
C27 is 7 uF at 13.5 V (datasheet curve).
U9: position for an open-drain voltage detector at the enable pin
of U10, not fitted. U10 has no soft start; as drawn it starts in
one go from the current-limited input (simulated; bench test open).
The 803 type, time-out 0.24 s, is no longer the part meant: a
remedy, if the bench asks for one, has a time-out of milliseconds.
R37
0.47R 1% 0.25W
PWR_FLAG
C29
10u 25V X7R
U13
LT3042EMSE
IN
EN/UV
ILIM
SET
OUT
OUTS
PGFB
PG
R39
442k 1%
R40
12.4k 1%
C30
4.7u 25V X7R
R38
120k 0.1%
+12V_A
TP16
C31
10u 25V X7R
D9
B0530W 30V 0.5A
R37 damps the capacitors on both sides and filters the ripple: 49 kHz with C29.
U13: VOUT = 100 uA x R38 = 12.0 V; enabled by 5V_OK (1.24 V rising).
R39, R40 end the fast charge of C30 at 11 V; settled after 0.8 s (calculated).
D9 clamps +12V_A near ground while −4V_A is present without it.
R30
2.2R 1% 0.25W
C21
22u 10V X7R
U11 LM27761DSG
VIN
EN
C+
C−
VOUT
VFB
CPOUT
GND
PAD
TP12
−4V_A
5V_OK
R27
100k 1%
R28
150k 1%
C19
1u 25V X7R
C22
4.7u 25V X7R
R34
226k 0.1%
R36
100k 0.1% 25ppm
C28
4.7u 25V X7R
D8 B0530W 30V 0.5A
U11: VOUT = −1.22 V x (R34 + R36) / R36 = −3.98 V.
R30, C21 keep fast edges of +5V off the 5.8 V rating of VIN.
R27, R28: EN = 0.6 x 5V_OK, below VIN while U11 limits its
input current. D8 clamps −4V_A to ground while U11 is off.
Links for the first power-up: L1, R37, R31 and R27;
each starts one stage.
+3V3_A
R31
10R 1%
C20
4.7u 25V X7R
U12
Vin
Vout
Temp
GND
Trim/NR
REF5025AID
VREF
TP14
VREF
D7
B0530W 30V 0.5A
R35
1R 1%
C24
1u 25V X7R
C26
10u 16V X7R
U12 is supplied from +3V3_A through R31 and C20
(3.9 kHz), so VREF is never present without the
rail of the parts that receive it.
D7 empties the capacitors on VREF into +3V3_A
when that rail falls. R35 damps C26; it is a
tuning position, 0 ohm to 1.5 ohm.
C24: noise filter, 10 Hz to 20 Hz (datasheet).
PWR_FLAG
+13V5
+13.5 V boost, +12 V_A, −4 V_A and the 2.5 V reference
Sheet: /Analog Rails/
File: analog_rails.kicad_sch
Title: Analog Rails and Reference
Size: A4
Date: 2026-10-09
KiCad E.D.A. 10.0.0
Rev: A2 (draft)
Id: 5/15



PWR_GOOD: four open-drain comparators share the pull-up R51, so the line is high only while all four rails are present. R41, R43 and R44 set the thresholds: 0.9001 x +3V3_C = 2.970 V and 0.4985 x +3V3_C = 1.645 V. Trip levels (calculated): +3V3_A 2.97 V, +12V_A 9.85 V, −4V_A −2.57 V, VREF 2.245 V. High level 0.82 x +3V3_C (R51 with R52); R52 holds the line low without supply. Start (simulated): 5V_OK enables +3V3_C and +3V3_A, then −4V_A (0.3 ms), +12V_A (8 ms to 17 ms), VREF and PWR_GOOD (about 24 ms). +13V5 is up before 5V_OK.

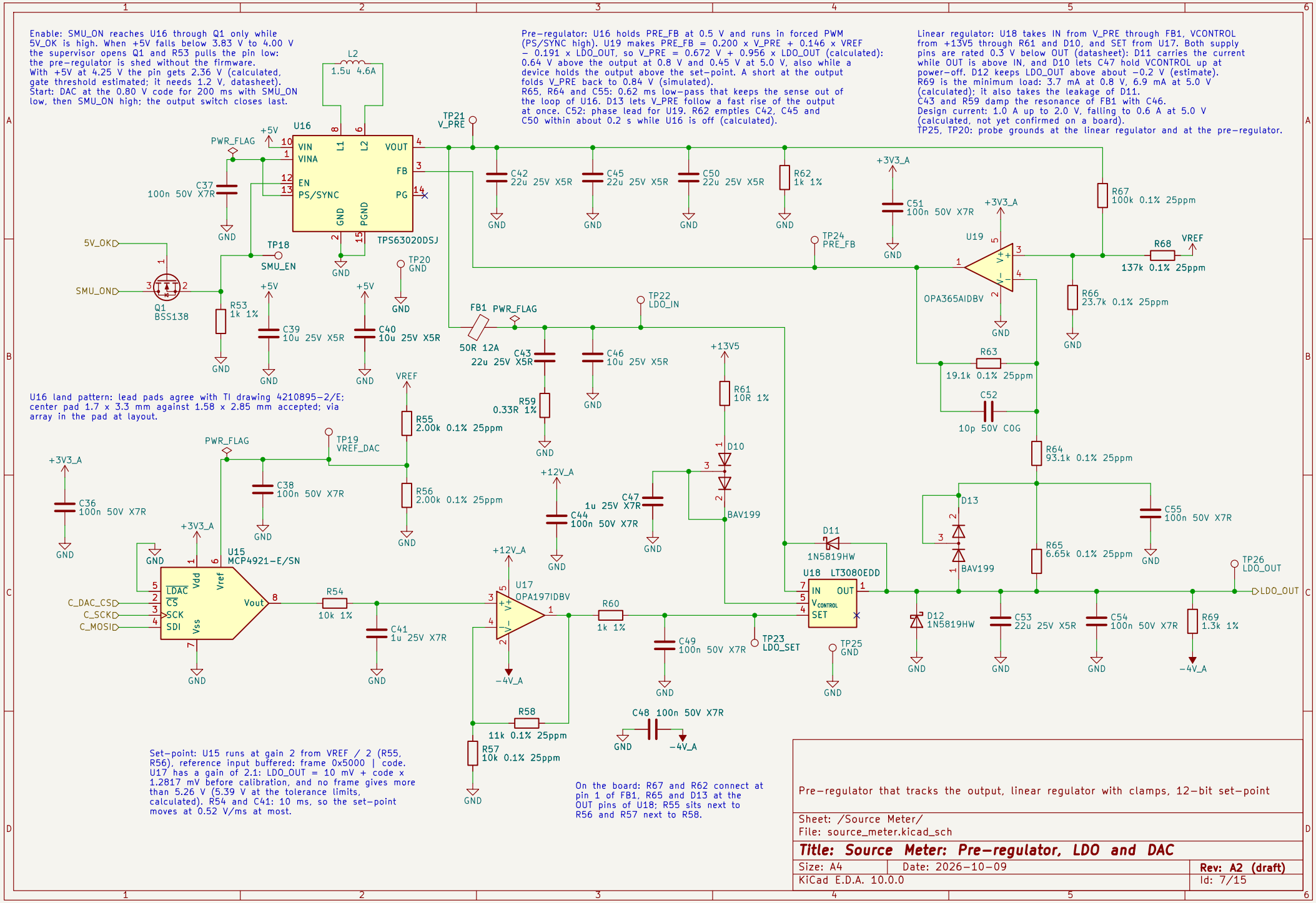
C32, C33 and C34, 10 nF each, keep an edge of PWR_GOOD from moving the thresholds through the pin capacitance of the package: each output pin of U14 is the neighbor of an inverting input (time constants about 0.09 ms, 0.25 ms and 0.36 ms, calculated).

Four comparators watch the analog rails and the reference; PWR_GOOD is their wired AND

Sheet: /Rail Monitor/
File: rail_monitor.kicad_sch

Title: Rail Monitor: PWR_GOOD

Size: A4	Date: 2026-10-09	**Rev: A2 (draft)**
KiCad E.D.A. 10.0.0		Id: 6/15


Enable: SMU_ON reaches U16 through Q1 only while 5V_OK is high. When +5V falls below 3.83 V to 4.00 V the supervisor opens Q1 and R53 pulls the pin low: the pre-regulator is shed without the firmware. With +5V at 4.25 V the pin gets 2.36 V (calculated, gate threshold estimated; it needs 1.2 V, datasheet). Start: DAC at the 0.80 V code for 200 ms with SMU_ON low, then SMU_ON high; the output switch closes last.
Pre-regulator: U16 holds PRE_FB at 0.5 V and runs in forced PWM (PS/SYNC high). U19 makes PRE_FB = 0.200 x V_PRE + 0.146 x VREF − 0.191 x LDO_OUT, so V_PRE = 0.672 V + 0.956 x LDO_OUT (calculated): 0.64 V above the output at 0.8 V and 0.45 V at 5.0 V, also while a device holds the output above the set-point. A short at the output folds V_PRE back to 0.84 V (simulated). R65, R64 and C55: 0.62 ms low-pass that keeps the sense out of the loop of U16. D13 lets V_PRE follow a fast rise of the output at once. C52: phase lead for U19. R62 empties C42, C45 and C50 within about 0.2 s while U16 is off (calculated).
Linear regulator: U18 takes IN from V_PRE through FB1, VCONTROL from +13V5 through R61 and D10, and SET from U17. Both supply pins are rated 0.3 V below OUT (datasheet): D11 carries the current while OUT is above IN, and D10 lets C47 hold VCONTROL up at power-off. D12 keeps LDO_OUT above about −0.2 V (estimate). R69 is the minimum load: 3.7 mA at 0.8 V, 6.9 mA at 5.0 V (calculated); it also takes the leakage of D11. C43 and R59 damp the resonance of FB1 with C46. Design current: 1.0 A up to 2.0 V, falling to 0.6 A at 5.0 V (calculated, not yet confirmed on a board). TP25, TP20: probe grounds at the linear regulator and at the pre-regulator.
U16 land pattern: lead pads agree with TI drawing 4210895-2/E; center pad 1.7 x 3.3 mm against 1.58 x 2.85 mm accepted; via array in the pad at layout.
Set-point: U15 runs at gain 2 from VREF / 2 (R55, R56), reference input buffered: frame 0x5000 | code. U17 has a gain of 2.1: LDO_OUT = 10 mV + code x 1.2817 mV before calibration, and no frame gives more than 5.26 V (5.39 V at the tolerance limits, calculated). R54 and C41: 10 ms, so the set-point moves at 0.52 V/ms at most.
On the board: R67 and R62 connect at pin 1 of FB1, R65 and D13 at the OUT pins of U18: R55 sits next to R56 and R57 next to R58.
L2
1.5u 4.6A
U16
TPS63020DSJ
VIN
VINA
L1
L2
VOUT
FB
EN
PS/SYNC
PG
GND
PGND
+5V
PWR_FLAG
C37
100n 50V X7R
TP21
V_PRE
C42
22u 25V X5R
C45
22u 25V X5R
C50
22u 25V X5R
R62
1k 1%
+3V3_A
C51
100n 50V X7R
R67
100k 0.1% 25ppm
5V_OK
SMU_ON
Q1
BSS138
TP18
SMU_EN
TP20
TP24
PRE_FB
U19
OPA365AIDBV
R68
137k 0.1% 25ppm
VREF
R66
23.7k 0.1% 25ppm
R53
1k 1%
C39
10u 25V X5R
C40
10u 25V X5R
FB1
50R 12A
TP22
LDO_IN
C43
22u 25V X5R
C46
10u 25V X5R
R59
0.33R 1%
R63
19.1k 0.1% 25ppm
C52
10p 50V C0G
+13V5
R61
10R 1%
D10
BAV199
R64
93.1k 0.1% 25ppm
R55
2.00k 0.1% 25ppm
TP19
VREF_DAC
C38
100n 50V X7R
R56
2.00k 0.1% 25ppm
+12V_A
C44
100n 50V X7R
C47
1u 25V X7R
D13
BAV199
C55
100n 50V X7R
C36
100n 50V X7R
U15
MCP4921-E/SN
LDAC
CS
SCK
SDI
Vdd
Vref
Vout
Vss
C_DAC_CS
C_SCK
C_MOSI
R54
10k 1%
C41
1u 25V X7R
U17
OPA197IDBV
R60
1k 1%
C49
100n 50V X7R
TP23
LDO_SET
D11
1N5819HW
U18
LT3080EDD
IN
OUT
VCONTROL
SET
R65
6.65k 0.1% 25ppm
TP26
LDO_OUT
D12
1N5819HW
C53
22u 25V X5R
C54
100n 50V X7R
R69
1.3k 1%
TP25
−4V_A
R58
11k 0.1% 25ppm
R57
10k 0.1% 25ppm
C48
100n 50V X7R
Pre-regulator that tracks the output, linear regulator with clamps, 12-bit set-point
Sheet: /Source Meter/
File: source_meter.kicad_sch
Title: Source Meter: Pre-regulator, LDO and DAC
Size: A4
Date: 2026-10-09
Rev: A2 (draft)
KiCad E.D.A. 10.0.0
Id: 7/15

Mode pairs: Q4, Q8 pass the regulator output (source mode), Q5, Q9 pass VIN (ampere mode). Both are open at reset. The gate networks are alike; for the ampere pair: Closing: R80 charges the gate against C61, R82: (100 k + 6.8 k) x 100 nF = 10.7 ms, after a first step of 0.76 V, below the threshold (calculated). The pair closes as a source follower: SUPPLY rises with about 0.9 V/ms, full gate drive in 40 ms (simulated). Opening: D17 (pin 1 to 3) and R79 discharge the gate at once: one pair is open before the other conducts. D17 (pin 3 to 2) bounds the gate at +12V_A plus two drops. Q7 joins gate and common source while that node is below ground: an open pair stays open with a reversed supply. D19 (pins 2 and 3 joined) holds the base one diode drop below the emitter otherwise. R84 returns the gate to the common source and keeps 95.7 % of the drive (calculated); no resistor goes from a gate to ground. TP32 and TP31 sit on 100 k nodes: a x10 probe lowers the gate by about 1 % (calculated). TP28, TP29: probe grounds at the two mode pairs.

Requests: R70 and R71 hold GATE_SRC and GATE_AMP low while the controller pins float. Q2 holds the ampere request low at U20 while source mode is requested, Q3 while VIN_OV is high: the two pairs are never closed together, whatever the controller asks. R75 and R72 limit the current into U20 while +12V_A is absent (2.7 mA, calculated). Firmware leaves 5 ms between lowering one request and raising either; a pair counts as closed 40 ms after its request, and the output switch closes after the mode pair.

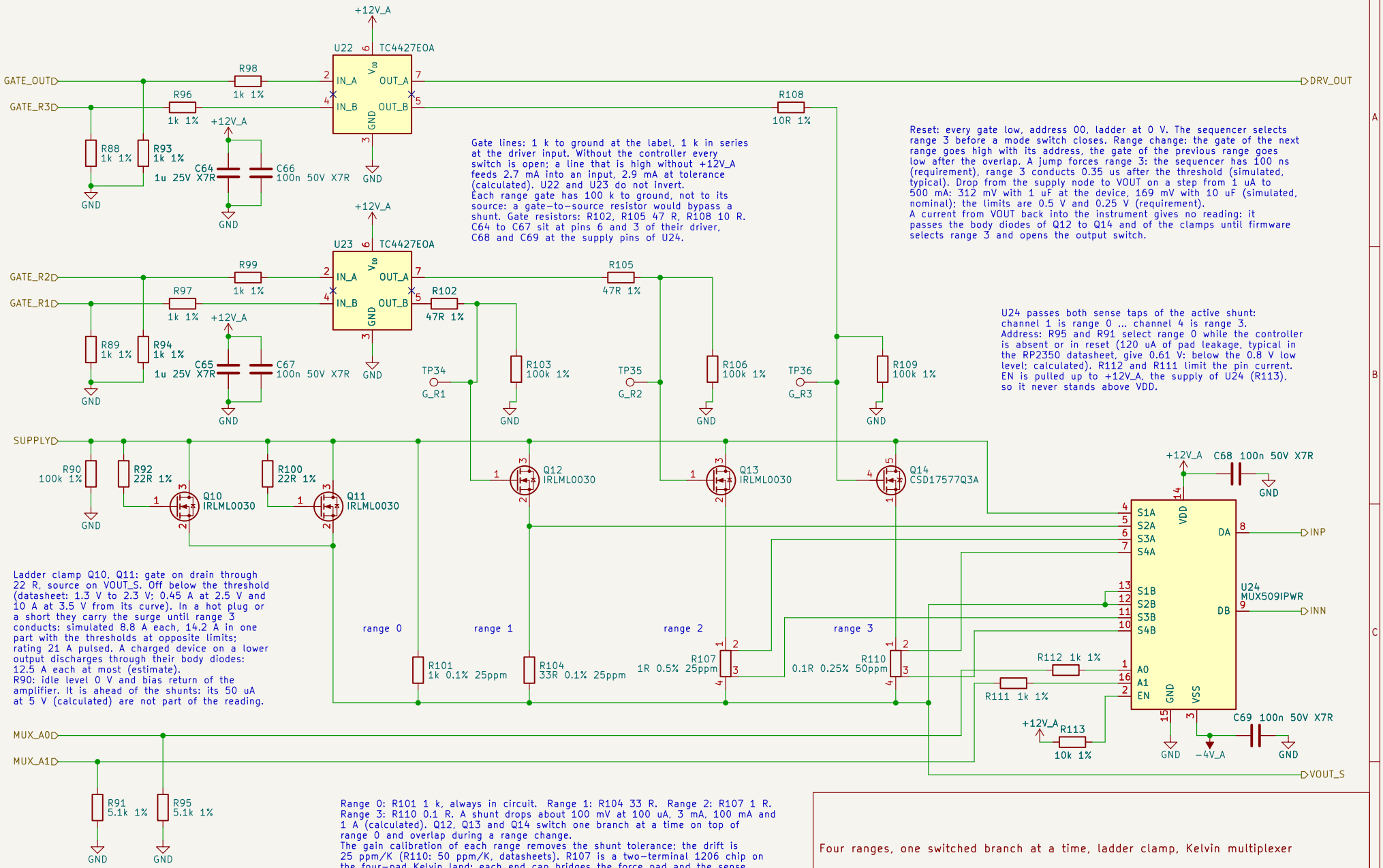
+12V_A
D16
R77 22R 1%
R78 100k 1%
BAV199
C60 100n 50V X7R
R81 6.8k 1%
TP28 GND
R83 2.2M 1%
D18
R85 100k 1%
Q6 BC847B
TP31 SRC_G
LDO_OUT
Q4 CSD17577Q3A
Q8 CSD17577Q3A
TP33 SUPPLY
C62 1u 25V X7R
R87 0.47R 1% 0.5W
C63 4.7u 25V X7R
SUPPLY
C56 1u 25V X7R
C57 100n 50V X7R
U20 TC4427EOA
GATE_SRC
GATE_AMP
R75 1k 1%
R72 1k 1%
R70 1k 1%
R71 4.7k 1%
Q2 BSS138
Q3 BSS138
D17
R79 22R 1%
R80 100k 1%
C61 100n 50V X7R
R82 6.8k 1%
R84 2.2M 1%
D19
R86 100k 1%
Q7 BC847B
TP32 AMP_G
Q5 CSD17577Q3A
Q9 CSD17577Q3A
Supply node of the ladder. C62 and the damped branch R87, C63 take the current of the supply leads when the over-current trip opens the output in ampere mode. Simulated: SUPPLY stays 0.5 V or more below +12V_A then. The bleed resistor of this node is on the Shunt Ladder sheet.
TP27 VIN_P
VIN_RAW
F1 4A fast 32V
D14 SMAJ20CA
+3V3_A
D15 BAV199
R73 115k 0.1%
R74 100k 0.1% 25ppm
R76 3.3M 1%
C58 1n 50V C0G
U21 MCP6561T-E/OT
VREF
C59 100n 50V X7R
TP30 VIN_OV
VIN_P
VIN_OV
VIN: 0.8 V to 5.0 V in use. With the ampere pair open the terminal withstands −20 V to +20 V and takes less than 1 mA (simulated). With the pair closed the supply has to stay in its range. D14 takes hot-plug rings and the kick of the supply leads; it conducts from 22.2 V (datasheet): never a 24 V supply. F1 does not protect D14. Detector: VIN_OV goes high at 5.46 V and low again at 5.35 V (calculated; 2.5 V at the tap of R73 and R74), 5.41 V to 5.51 V and 5.30 V to 5.40 V with the tolerances. Not latched. D15 keeps the input within a diode drop of both rails; R73 limits its current to 0.17 mA at 20 V (calculated).

Source and ampere mode switches with their gate networks, VIN input and its protection

Sheet: /Path Switching/
File: path_switching.kicad_sch

Title: Mode Switches and VIN Protection

Size: A4 | Date: 2026-10-09 | **Rev: A2 (draft)**
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Gate lines: 1 k to ground at the label, 1 k in series at the driver input. Without the controller every switch is open; a line that is high without +12V_A feeds 2.7 mA into an input, 2.9 mA at tolerance (calculated). U22 and U23 do not invert.
Each range gate has 100 k to ground, not to its source: a gate-to-source resistor would bypass a shunt. Gate resistors: R102, R105 47 R, R108 10 R. C64 to C67 sit at pins 6 and 3 of their driver, C68 and C69 at the supply pins of U24.

Reset: every gate low, address 00, ladder at 0 V. The sequencer selects range 3 before a mode switch closes. Range change: the gate of the next range goes high with its address, the gate of the previous range goes low after the overlap. A jump forces range 3: the sequencer has 100 ns (requirement), range 3 conducts 0.35 us after the threshold (simulated, typical). Drop from the supply node to VOUT on a step from 1 uA to 500 mA: 312 mV with 1 uF at the device, 169 mV with 10 uF (simulated, nominal); the limits are 0.5 V and 0.25 V (requirement).
A current from VOUT back into the instrument gives no reading: it passes the body diodes of Q12 to Q14 and of the clamps until firmware selects range 3 and opens the output switch.

U24 passes both sense taps of the active shunt: channel 1 is range 0 ... channel 4 is range 3. Address: R95 and R91 select range 0 while the controller is absent or in reset (120 uA of pad leakage, typical in the RP2350 datasheet, give 0.61 V: below the 0.8 V low level; calculated). R112 and R111 limit the pin current. EN is pulled up to +12V_A, the supply of U24 (R113), so it never stands above VDD.

Ladder clamp Q10, Q11: gate on drain through 22 R, source on VOUT_S. Off below the threshold (datasheet: 1.3 V to 2.3 V; 0.45 A at 2.5 V and 10 A at 3.5 V from its curve). In a hot plug or a short they carry the surge until range 3 conducts: simulated 8.8 A each, 14.2 A in one part with the thresholds at opposite limits; rating 21 A pulsed. A charged device on a lower output discharges through their body diodes: 12.5 A each at most (estimate).
R90: idle level 0 V and bias return of the amplifier. It is ahead of the shunts: its 50 uA at 5 V (calculated) are not part of the reading.

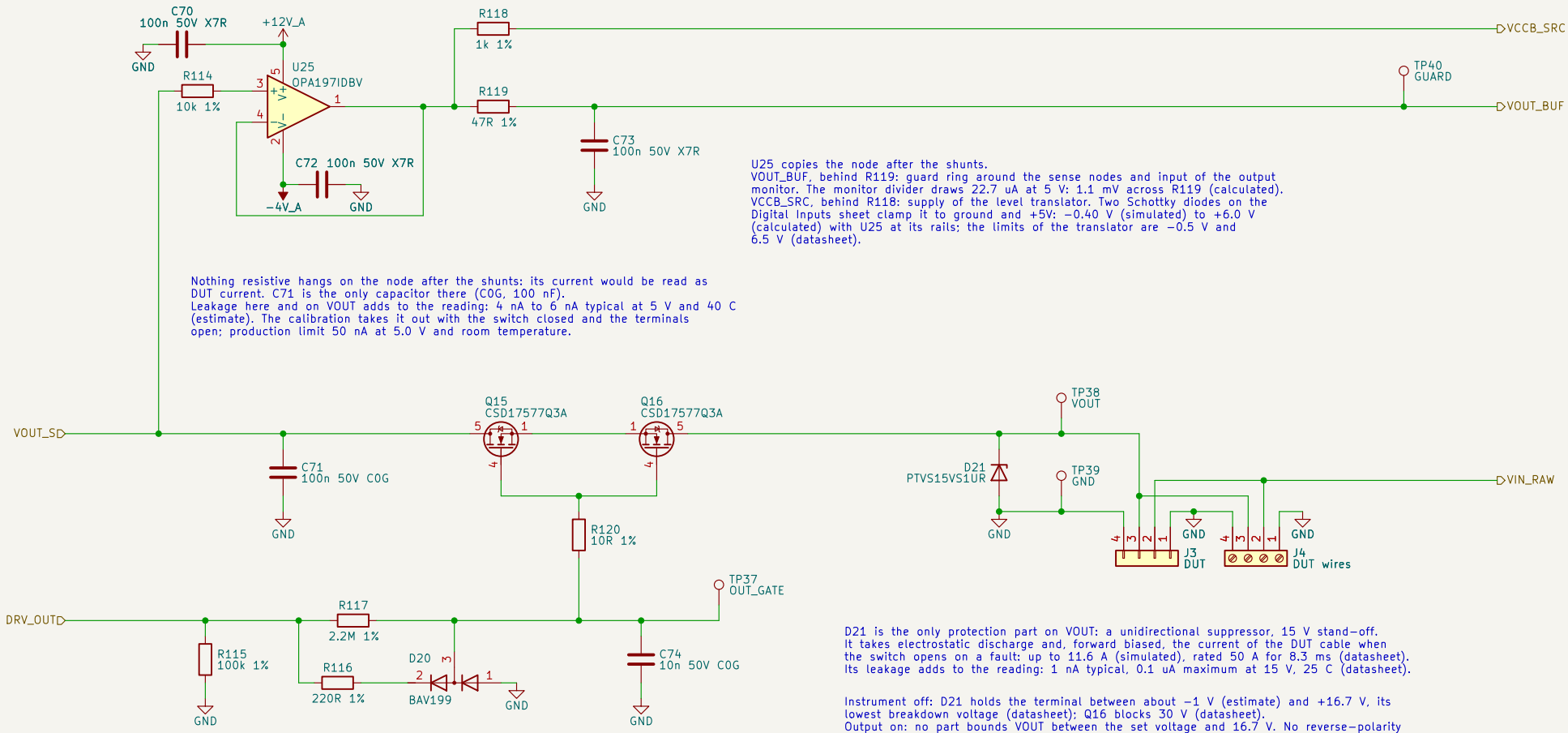
Range 0: R101 1 k, always in circuit. Range 1: R104 33 R. Range 2: R107 1 R. Range 3: R110 0.1 R. A shunt drops about 100 mV at 100 uA, 3 mA, 100 mA and 1 A (calculated). Q12, Q13 and Q14 switch one branch at a time on top of range 0 and overlap during a range change.
The gain calibration of each range removes the shunt tolerance; the drift is 25 ppm/K (R110: 50 ppm/K, datasheets). R107 is a two-terminal 1206 chip on the four-pad Kelvin land: each end cap bridges the force pad and the sense pad of its end. R110 is a four-terminal part. Ranges 0 and 1 share their force and sense nets: route their sense tracks from the pads.
Leakage budget across the ladder: 100 nA at 100 mV and 40 C (requirement); the clamp type is a candidate.

Four ranges, one switched branch at a time, ladder clamp, Kelvin multiplexer

Sheet: /Shunt Ladder/
File: shunt_ladder.kicad_sch

Title: Shunt Ladder and Kelvin Multiplexer

Size: A4	Date: 2026-10-09	**Rev: A2 (draft)**
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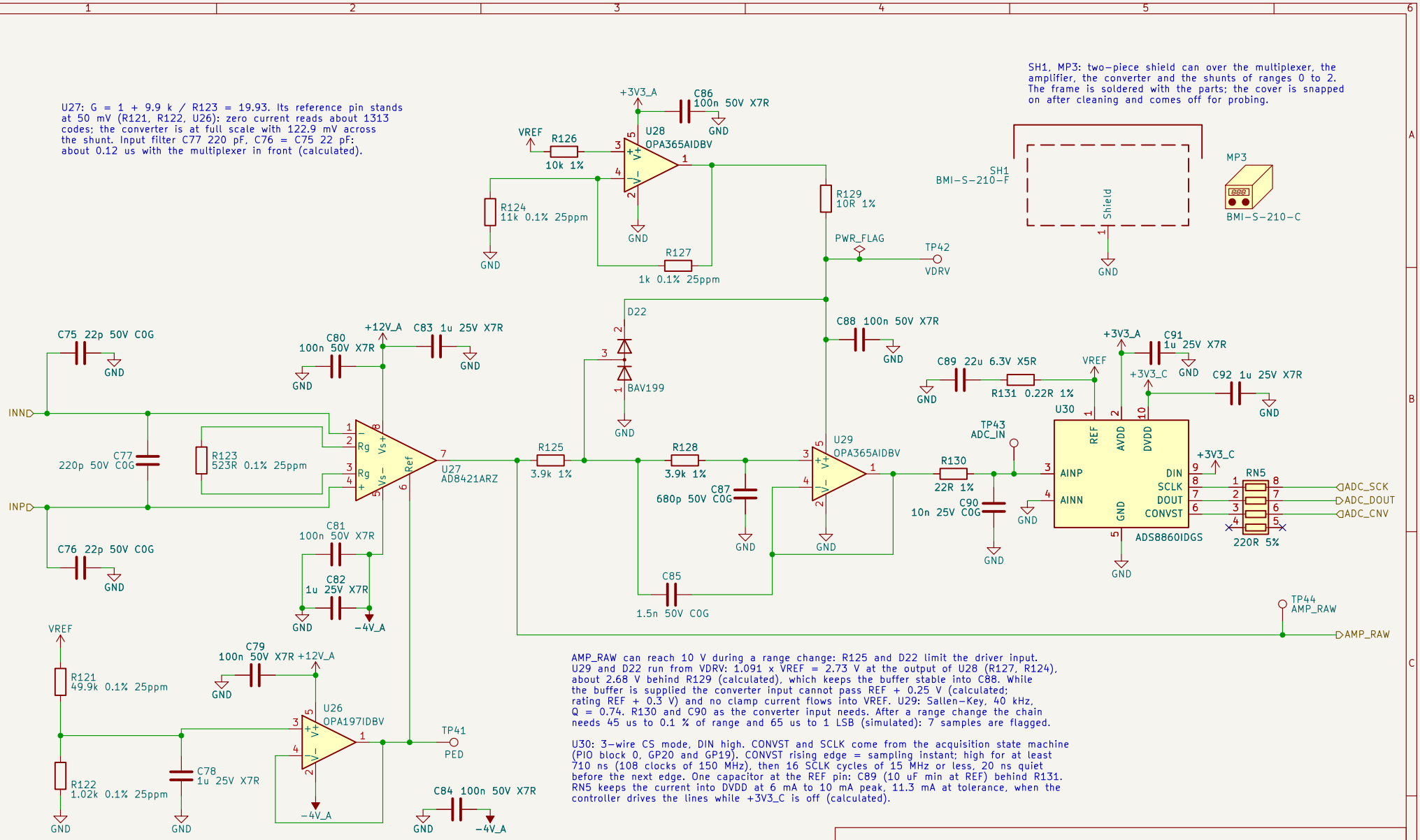


Output switch with slow turn−on, DUT connectors with their protection, guard buffer

Sheet: /Output Stage/
File: output_stage.kicad_sch

Title: Output Switch, DUT Connectors and Guard Buffer

Size: A4	Date: 2026−10−09	Rev: A2 (draft)
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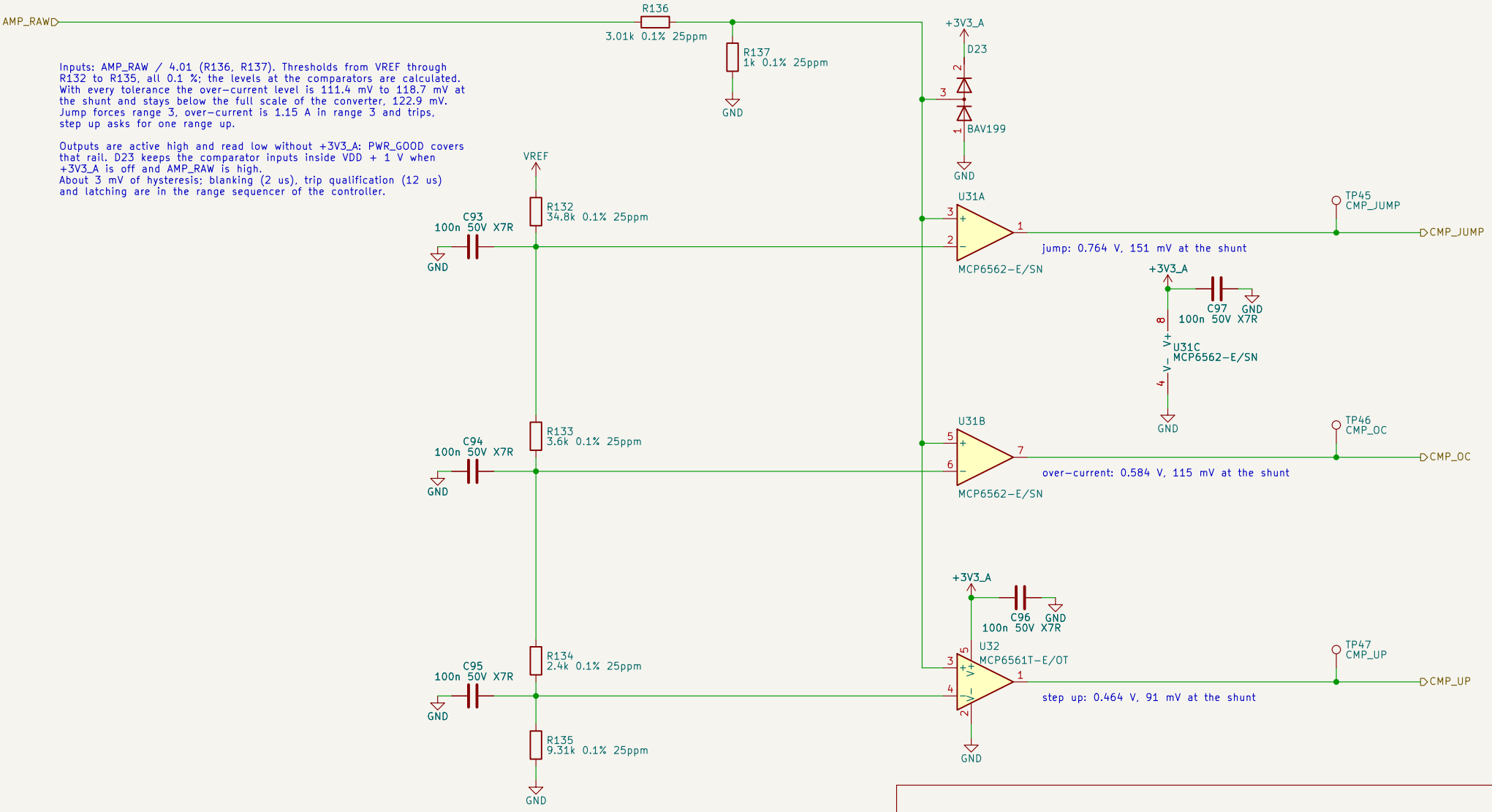


Gain 19.93, 50 mV pedestal, limiter on VDRV, 40 kHz two-pole filter, 16-bit converter

Sheet: /Signal Chain/
File: signal_chain.kicad_sch

Title: Amplifier, Anti-alias Filter and ADC

Size: A4	Date: 2026-10-09	**Rev: A2 (draft)**
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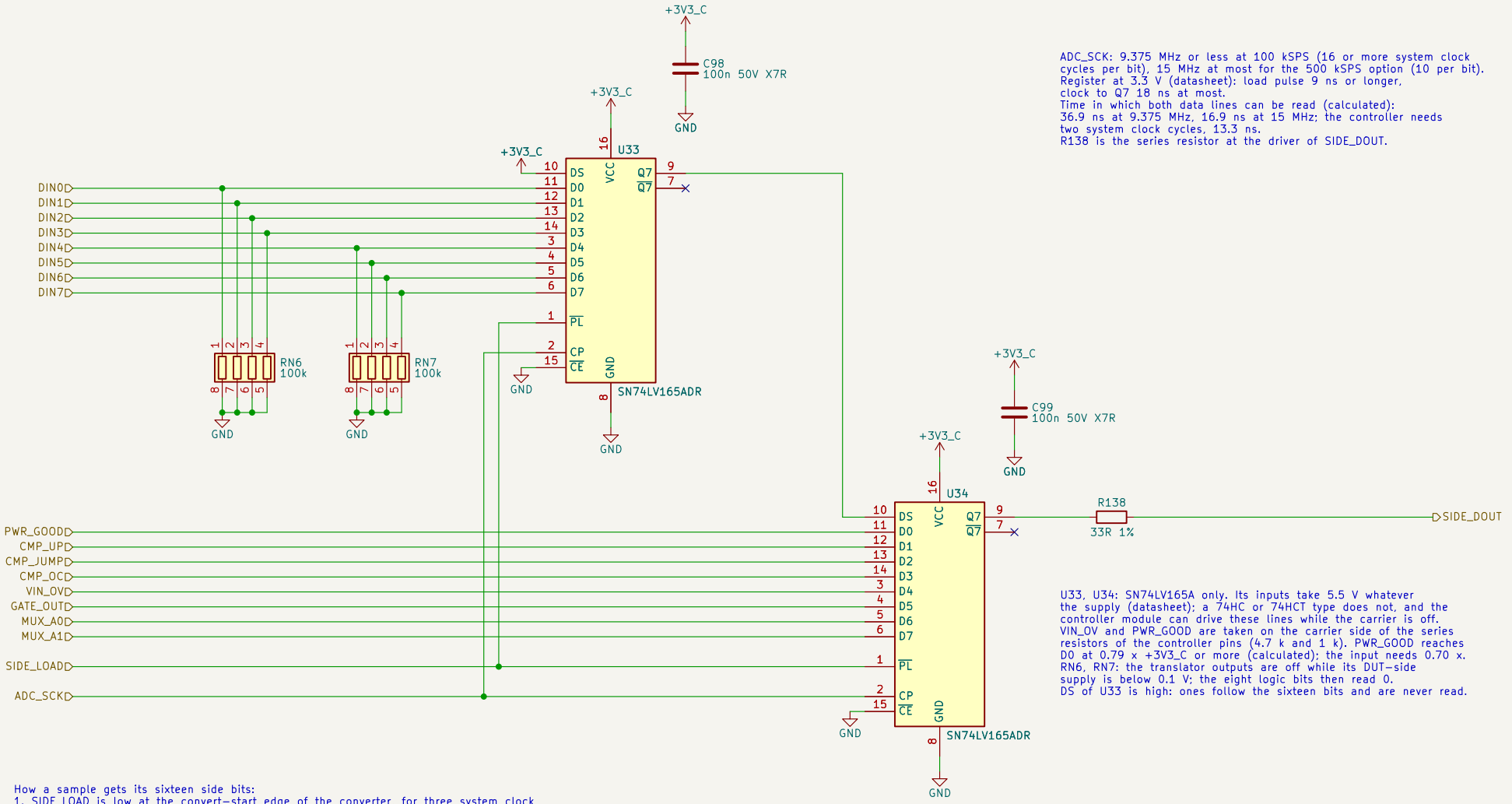


Step up at 91 mV, over-current at 115 mV, jump to range 3 at 151 mV at the shunt

Sheet: /Comparators/
File: comparators.kicad_sch

Title: Range and Over-current Comparators

Size: A4	Date: 2026-10-09	**Rev: A2 (draft)**
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ADC_SCK: 9.375 MHz or less at 100 kSPS (16 or more system clock cycles per bit), 15 MHz at most for the 500 kSPS option (10 per bit).
Register at 3.3 V (datasheet): load pulse 9 ns or longer, clock to Q7 18 ns at most.
Time in which both data lines can be read (calculated): 36.9 ns at 9.375 MHz, 16.9 ns at 15 MHz; the controller needs two system clock cycles, 13.3 ns.
R138 is the series resistor at the driver of SIDE_DOUT.

U33, U34: SN74LV165A only. Its inputs take 5.5 V whatever the supply (datasheet); a 74HC or 74HCT type does not, and the controller module can drive these lines while the carrier is off.
VIN_OV and PWR_GOOD are taken on the carrier side of the series resistors of the controller pins (4.7 k and 1 k). PWR_GOOD reaches D0 at 0.79 x +3V3_C or more (calculated); the input needs 0.70 x.
RN6, RN7: the translator outputs are off while its DUT-side supply is below 0.1 V; the eight logic bits then read 0.
DS of U33 is high: ones follow the sixteen bits and are never read.

How a sample gets its sixteen side bits:
1. SIDE_LOAD is low at the convert-start edge of the converter, for three system clock cycles (20 ns) or more. U33 and U34 follow their inputs while it is low and hold them when it rises; SIDE_DOUT then shows MUX_A1.
2. Each rising edge of ADC_SCK brings the next bit. The order on SIDE_DOUT: MUX_A1, MUX_A0 (range), GATE_OUT, VIN_OV, CMP_OC, CMP_JUMP, CMP_UP, PWR_GOOD, then the logic inputs DIN7 down to DIN0.
3. The controller reads SIDE_DOUT and the data pin of the converter at the same instants: every sample carries the range, the flags and the logic levels of its own instant.
Controller in reset or absent: pull-downs on the Controller sheet hold SIDE_LOAD and ADC_SCK low; the registers follow their inputs and nothing is shifted.

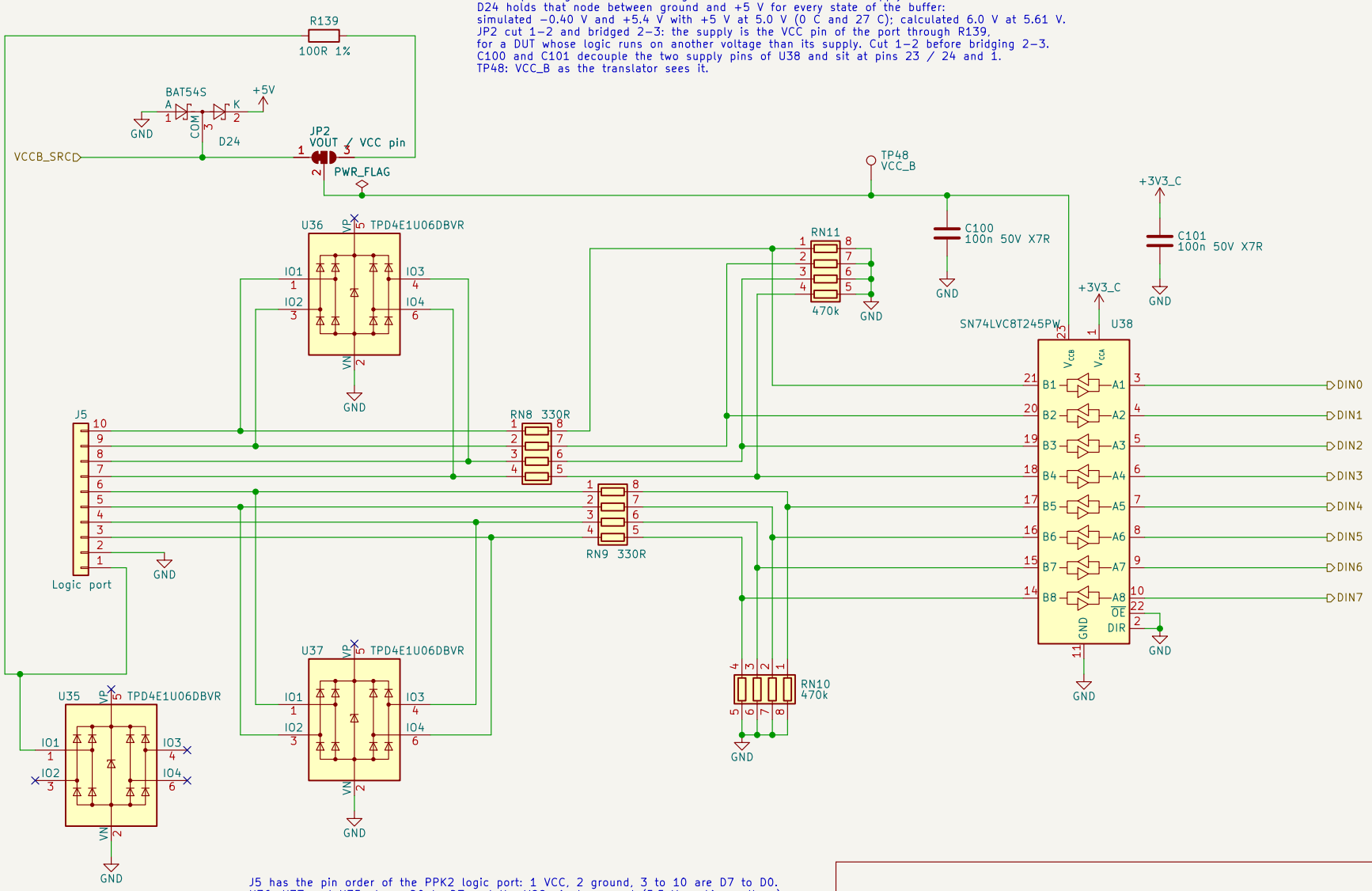
Sixteen bits captured with every sample: status first, then the eight logic inputs

Sheet: /Side Data/
File: side_data.kicad_sch

Title: Side Data: Status and Logic Inputs of Every Sample

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Supply of port B of U38: 1.65 V to 5.5 V in use, limits −0.5 V and 6.5 V (datasheet).
JP2 bridged 1–2 as built: the supply is VCCB_SRC, the guard buffer output behind 1 k on the Output Stage sheet, so the logic thresholds follow the DUT supply.
D24 holds that node between ground and +5 V for every state of the buffer: simulated −0.40 V and +5.4 V with +5 V at 5.0 V (0 C and 27 C); calculated 6.0 V at 5.61 V.
JP2 cut 1–2 and bridged 2–3: the supply is the VCC pin of the port through R139, for a DUT whose logic runs on another voltage than its supply. Cut 1–2 before bridging 2–3.
C100 and C101 decouple the two supply pins of U38 and sit at pins 23 / 24 and 1.
TP48: VCC_B as the translator sees it.



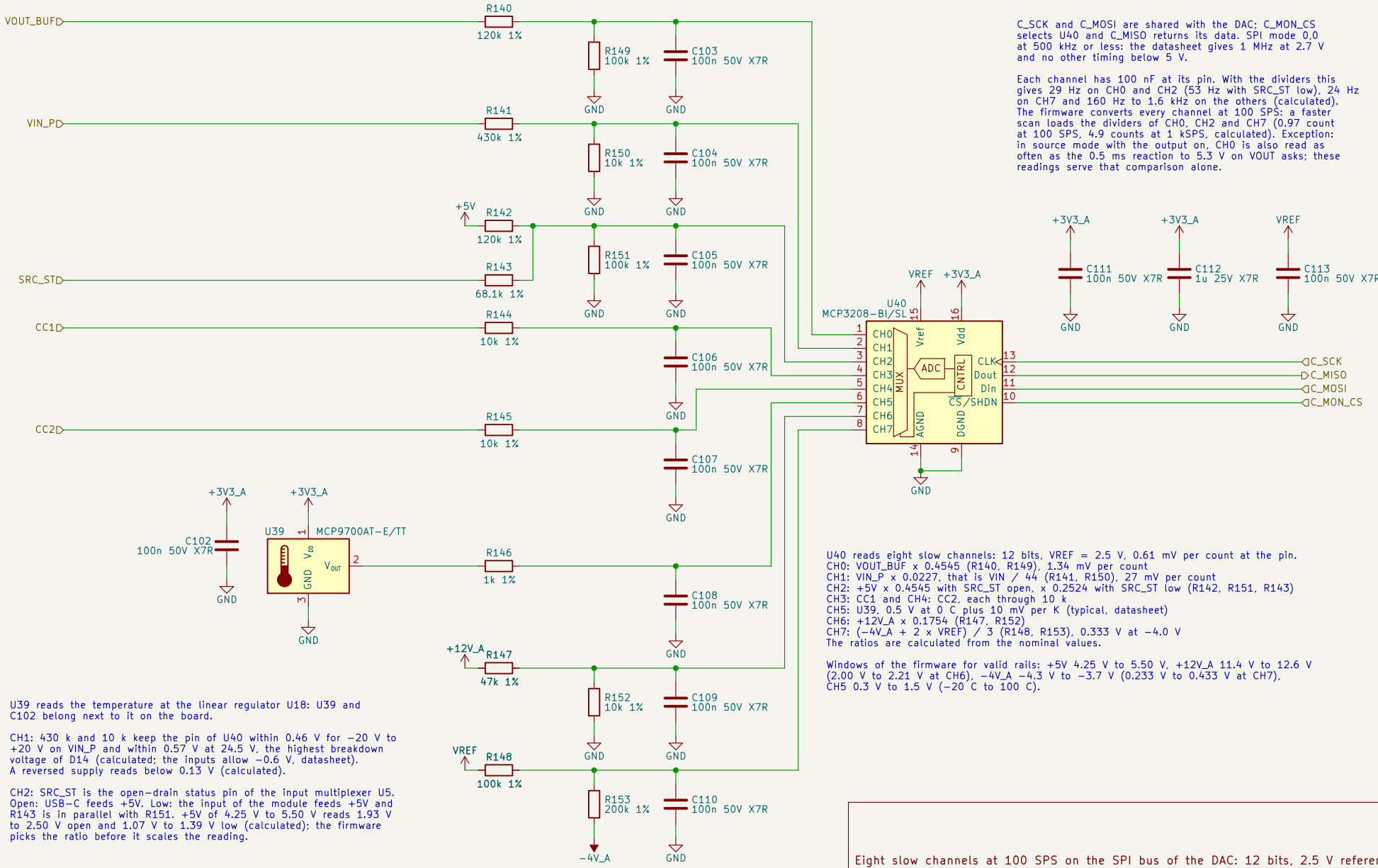
J5 has the pin order of the PPK2 logic port: 1 VCC, 2 ground, 3 to 10 are D7 to D0.
U36, U37 and U35 clamp D0 to D7 and the VCC pin to ground (5.5 V working voltage).
RN8 / RN9 limit the input current and keep the edge slope at U38 inside its limit (calculated 2.6 to 3.3 ns/V at 5 V against 5 ns/V). RN11 / RN10 hold an open input low for a pin leakage up to 1 uA, the 25 C limit of the datasheet. A line held high loads the DUT with V / 470 k: 3.5 uA at 1.65 V, 7 uA at 3.3 V, 10.6 uA at 5 V.
DIR and OE are tied low: enabled, direction B to A. With port B unpowered the outputs float; pull-downs on the Side Data sheet hold DIN0 to DIN7.

Logic port in the pin order of the PPK2; the inputs follow the DUT logic levels

Sheet: /Digital Inputs/
File: digital_inputs.kicad_sch

Title: Logic Port: Digital Inputs D0 to D7

Size: A4	Date: 2026-10-09	**Rev: A2 (draft)**
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C_SCK and C_MOSI are shared with the DAC; C_MON_CS selects U40 and C_MISO returns its data. SPI mode 0,0 at 500 kHz or less: the datasheet gives 1 MHz at 2.7 V and no other timing below 5 V.

Each channel has 100 nF at its pin. With the dividers this gives 29 Hz on CH0 and CH2 (53 Hz with SRC_ST low), 24 Hz on CH7 and 160 Hz to 1.6 kHz on the others (calculated). The firmware converts every channel at 100 SPS: a faster scan loads the dividers of CH0, CH2 and CH7 (0.97 count at 100 SPS, 4.9 counts at 1 kSPS, calculated). Exception: in source mode with the output on, CH0 is also read as often as the 0.5 ms reaction to 5.3 V on VOUT asks; these readings serve that comparison alone.

U40 reads eight slow channels: 12 bits, VREF = 2.5 V, 0.61 mV per count at the pin.
CH0: VOUT_BUF x 0.4545 (R140, R149), 1.34 mV per count
CH1: VIN_P x 0.0227, that is VIN / 44 (R141, R150), 27 mV per count
CH2: +5V x 0.4545 with SRC_ST open, x 0.2524 with SRC_ST low (R142, R151, R143)
CH3: CC1 and CH4: CC2, each through 10 k
CH5: U39, 0.5 V at 0 C plus 10 mV per K (typical, datasheet)
CH6: +12V_A x 0.1754 (R147, R152)
CH7: (−4V_A + 2 x VREF) / 3 (R148, R153), 0.333 V at −4.0 V
The ratios are calculated from the nominal values.

Windows of the firmware for valid rails: +5V 4.25 V to 5.50 V, +12V_A 11.4 V to 12.6 V (2.00 V to 2.21 V at CH6), −4V_A −4.3 V to −3.7 V (0.233 V to 0.433 V at CH7), CH5 0.3 V to 1.5 V (−20 C to 100 C).

U39 reads the temperature at the linear regulator U18: U39 and C102 belong next to it on the board.

CH1: 430 k and 10 k keep the pin of U40 within 0.46 V for −20 V to +20 V on VIN_P and within 0.57 V at 24.5 V, the highest breakdown voltage of D14 (calculated; the inputs allow −0.6 V, datasheet). A reversed supply reads below 0.13 V (calculated).

CH2: SRC_ST is the open−drain status pin of the input multiplexer U5. Open: USB−C feeds +5V. Low: the input of the module feeds +5V and R143 is in parallel with R151. +5V of 4.25 V to 5.50 V reads 1.93 V to 2.50 V open and 1.07 V to 1.39 V low (calculated); the firmware picks the ratio before it scales the reading.

Eight slow channels at 100 SPS on the SPI bus of the DAC: 12 bits, 2.5 V reference

Sheet: /Monitors/
File: monitors.kicad_sch

Title: Slow Monitors

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